SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
(Common to all Branches)

Mode of Exam
OFFLINE

DEPARTMENT OF COMPUTING TECHNOLOGIES
SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu

Academic Year: 2023-24 (ODD)

RegNo

Test: CLAT3 **Date: 8.11.2023**
Course Code & Title: 21CSS201T / COA **Duration: 100 minutes**
Year & Sem: II & III **SET D** **Max. Marks: 50**

Course Articulation Matrix:

Course Learning Outcomes (CLO)	At the end of this course, learners will be able to:	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO-4	Analyze concepts of parallelism and multi-core processors	H	-	-	-	-	-	-	-	-	-	-	-
CO-5	Classify the memory technologies, input-output systems and evaluate the performance of memory system	H	M	-	-	-	-	-	-	-	-	-	-

Part – A
Instructions: Answer all (10 x 1 = 10 Marks)

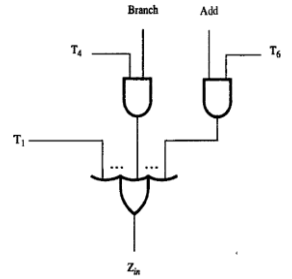
Q. No	Question	Marks	BL	CO	PO	PI Code
1.	A word whose individual bits represent a control signal is _____ a) Command word b) Control word c) Control status d) Generation word	1	1	4	1.3	1.3.1
2	Which of the following statement is true with respect to the function of WMFC signal? a. Used to make processor wait for the reply from memory b. Execution of memory instructions c. Idle state of memory d. No operation	1	2	4	1.3	1.3.1
3	Individual control words of micro routine are called as _____. a. Micro task b. Micro operations c. Micro instruction d. Micro commands	1	1	4	1.3	1.3.1
4	Let there be two instructions I1 and I2 such that: I1 : ADD R1, R2, R3 I2 : MUL R4, R1, R2 Identify the type of hazard which occurs in the above instruction A. Structural hazard B. Data hazard C. Control hazard D. Instructional hazard	1	1	4	1.3	1.3.1
5	A technique which is used to minimize the penalty caused by conditional branch is _____ A. Delayed branching B. Branch penalty	1	1	4	1.3	1.3.1

	C. Delay D. Prediction					
6	Which of the following is NOT a common challenge in parallel computing? A. Load balancing B. Scalability C. Sequential execution D. Data consistency	1	1	5	1	2.1.2
7	Which type of architecture is often used in vector processors, where operations are performed in parallel on elements of arrays or vectors? A. SISD B. SIMD C. MIMD D. MISD	1	1	5	1	2.1.2
8	In ARM assembly language, what instruction is used for storing data from a register into memory? A. LDR B. STR C. MOV D. ADD	1	1	5	1	2.1.2
9	Which ARM architecture is commonly used in embedded systems, such as microcontrollers and IoT devices? a. ARM5 b. ARM6 c. ARM7 d. ARM Cortex-M	1	1	5	1	2.1.2
10	In the MIPS32 pipeline, which of the following scenarios of data dependency will always result in a pipeline stall due to data hazard without any instruction scheduling? (a) An ADD instruction followed by a SUB instruction. (b) A STORE instruction followed by a LOAD instruction (c) A LOAD instruction followed by an ADD instruction. (d) None of the above.	1	1	5	1	2.1.2

Part – B
Instructions: Answer any 4 (4 x 4 = 16 Marks)

11 Draw the generation of Zin control signals for a single bus structure.

- **Z_{IN} = T1 + T6.ADD +T4.Br+…..**



12 Outline the limitations of Instruction Level Parallelism?

Dependency Chains

Branch Prediction Misprediction: Speculative execution relies on accurate branch prediction. Mispredictions can lead to wasted cycles and performance degradation.

Code Size: Increasing ILP often requires longer instruction sequences, which can lead to larger code size, impacting cache and memory efficiency.

13 Why WMFC signal is needed when doing operations with memory?

The WMFC (Wait for Memory-Function-Completed) signal causes the processor waits for the arrival of the MFC signal. The MFC signal is set by the addressed device when the contents of the specified location have been read and are available on the data lines of the memory bus.

14 **There is an instruction pipeline with four stages. The stage delays for each stage is 5 nsec, 6 nsec, 11 nsec, and 8 nsec respectively. Consider the delay of an inter-stage register in the pipeline is 1 nsec. Find the approximate speedup of the pipeline in the steady state under ideal conditions as compare to to the corresponding non-pipelined implementation?**

When N instruction will be execute in non pipeline manner the time taken will be equal be (5 + 6 + 11 + 8) * N = 30N

Now we will calculate the clock period for pipelined implementation and it will be equal to max{5,6,11,8} + 1 = 12.

Time taken in execution of N instruction through pipelined will be equal to 12N (approx.)

So Speedup = 30N / 12N = 2.5

15 **Summarize about Thumb instruction set.**

ARM architecture versions v4T and above define a 16-bit instruction set called the Thumb instruction set. The functionality of the Thumb instruction set is a subset of the functionality of the 32-bit ARM instruction set.

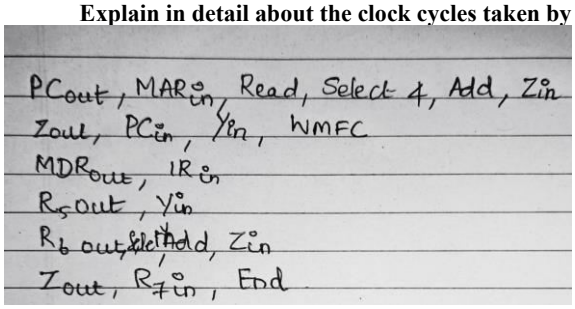
A processor that is executing Thumb instructions is operating in Thumb state. A processor that is executing ARM instructions is operating in ARM state.

A processor in ARM state cannot execute Thumb instructions, and a processor in Thumb state cannot execute ARM instructions. must ensure that the processor never receives instructions of the wrong instruction set for the current state. Each instruction set includes instructions to change processor state.

PART C

Instructions: Answer all (12 x 2 = 24 Marks)

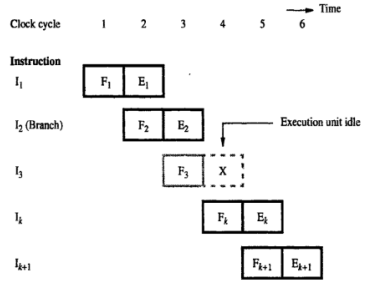
16. A **Compare the performance of the execution of the following set of instructions in single bus organizarion. Explain in detail about the clock cycles taken by both scenarios for the instruction ADD R5, R6, R7**



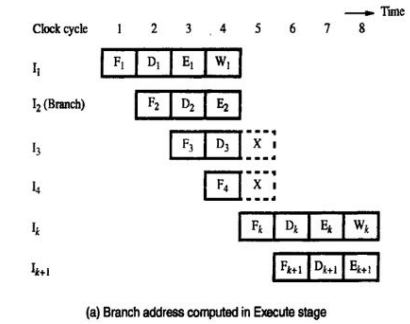
OR

16. B **Illustrate the pipelining for the 3 scenarios to be executed in 4 – stage pipelining.**

a) Idle stall for branch instruction

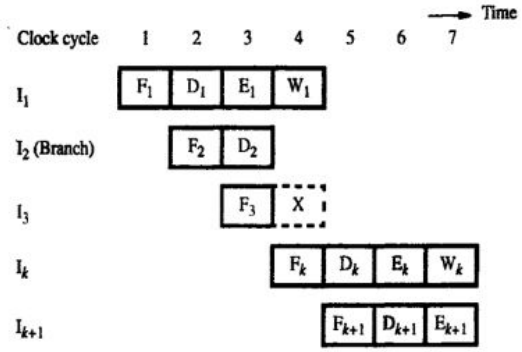


b) Branch predicted in execute stage



(a) Branch address computed in Execute stage

c) Branch predicted in decode stage



(b) Branch address computed in Decode stage

17 A. **You are a project manager overseeing the development of a new software application for a client. The project involves multiple teams and tasks that need to be completed in parallel. Your client has strict deadlines and expects a high-quality product. How would you use the concept of parallelism to ensure the project's success? Please provide a detailed plan**

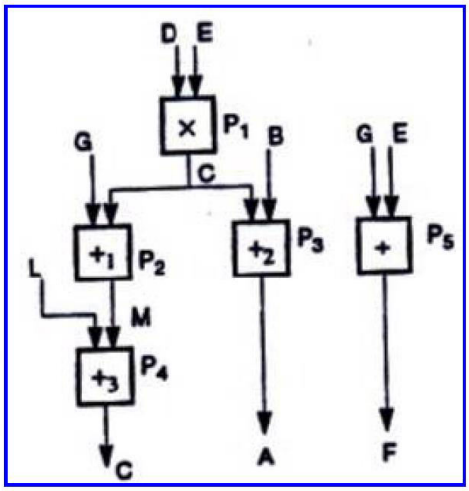
- Executing two or more operations at the same time is known as parallelism.
- Parallel processing is a method to improve computer system performance by executing two or more instructions simultaneously
- A parallel computer is a set of processors that are able to work cooperatively to solve a computational problem.
- Two or more ALUs in CPU can work concurrently to increase throughput
- The system may have two or more processors operating concurrently

Goals of parallelism

Types of parallelism

Parallel Execution:

$$
\begin{array}{lrcl}
P_1: & C & = & D \times E \\
P_2: & M & = & G + C \\
P_3: & A & = & B + C \\
P_4: & C & = & L + M \\
P_5: & F & = & G \div E
\end{array}
$$

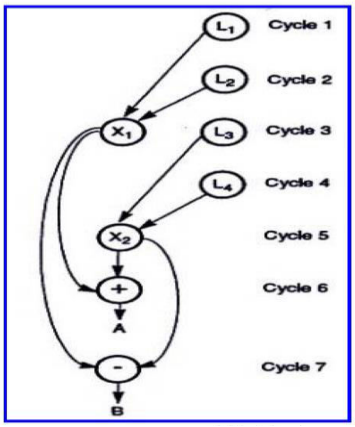


- Parallel execution in THREE steps, assuming TWO adders are available per step.
- If TWO adders are available, the parallel execution requires only THREE steps.
- Only 5 pairs can execute in parallel, if there is no resource conflicts.
- In this example, as shown in the fig, only P2 ‖ P3 ‖ P5 is possible.

Hardware Parallelism:

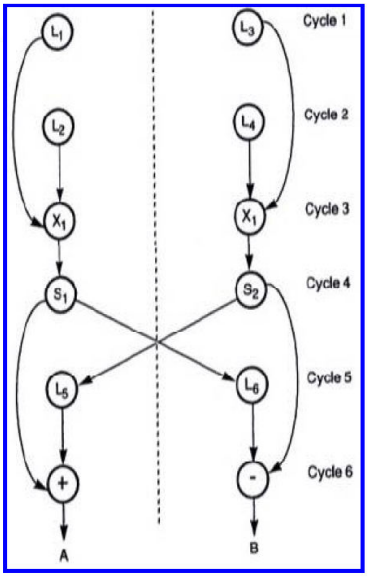
Parallel Execution:

- **Using TWO-issue processor:**
- **The processor can execute one memory access (Load or Store) and one arithmetic operation (multiply, add, subtract) simultaneously.**
- **The program must execute in 7 cycles.**
- **The h/w parallelism average is 8/7=1.14.**
- **It is clear from this example the mismatch between the s/w and h/w parallelism.**



Example:

- A h/w platform of a Dual-Processor system, single issue processors are used to execute the same program.
- Six processor cycles are needed to execute the 12 instructions by two processors.
- S1 & S2 are two inserted store operations.
- L5 & L6 are two inserted load operations.
- The added instructions are needed for inter-processor communication through the shared memory.



OR

17 B **Suppose you are a software architect working on a project to develop a real-time image processing system for a security camera network. You have two architectural choices: SISD or SIMD. Describe the advantages and disadvantages of each architecture in the context of this scenario, and explain which one you would recommend for the image processing system. Justify your recommendation**